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**UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS**

**International General Certificate of Secondary Education**

# MARK SCHEME for the October/November 2007 question paper

## 0625 PHYSICS

**0625/05** Paper 5 (Practical Test), maximum raw mark 40

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**1** **(a)–(e)** $t$ in s, $\theta$ in °C, and $\theta_0$ (10 – 45) [1]
Complete set of readings, temps decreasing [1]
Evidence of $\theta$ to 1°C [1]

**(f)** **(i)** $T_1$, $T_2$ correct arithmetic [1]

**(ii)** $T_1 > T_2$ [1]

**(g)** **(i)** reason consistent with results [1]

**(ii)** Three from:
room temp/draughts, etc.
volume/mass/amount
beaker/insulation/lid/surface
liquid
amount of stirring
(not starting temperature) [3]

**(h)** lid [1]

**[Total: 10]**

**2** **(a)** $h_0$ 25 – 100 cm with correct unit [1]

**(b)–(d)** complete table $h$, $d$ [1]
correct arithmetic for $d$ [1]
all $h$ to nearest mm [1]

**(e)** Graph:
suitable scale labelled symbol/unit [1]
all plots to nearest ½ sq (–1 each error or omission) [2]
line thin and well judged [1]

**(g)** calculation of $d$ correct [1]
correct reading from graph to ½ square and to 1dp [1]

**[Total: 10]**

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**3** **(a)–(c)** 4 $I$ values, sensible (watch for $I$ x 10) [1]
All $I$ to at least 2 dp [1]
$I$ in A at least once [1]
$I = I_1 + I_2 \pm I_3 + 10\%$ [1]

**(d)** statement (yes)
reason consistent with readings [1]

**(e)** variable resistor/extra cell/vary power supply/different number of lamps [1]

**(f)** sensible $V$ (< 3V), unit and at least 1 dp [1]

**(g)** correct arithmetic for $R$ [1]
unit and 2/3 sf [1]

**(h)** $V_a = 0$, $V_b = V$ [1]

**[Total: 10]**

**4** **(f)–(h)** sensible $x$ value (less than $h$) [1]
sensible $h$ value (typical block: 10 cm) [1]
$x$ to nearest mm [1]
$x$ and $h$ with same unit [1]

correct arithmetic for $n$ [1]

**(i)–(j)** second different $h$ value [1]

**(k)** correct method for average $n$ [1]
2/3 sf and no unit [1]
both $n$ values 1.4 – 1.6 [1]

**(l)** two equal heights from bench
(or other valid method) [1]

**[Total: 10]**

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